

QUILVIAN

Q-VOICE

Quilvian Patient Voice & Experience Intelligence

Patient Experience Score - Complaint & Service Recovery - Management Recommendation - Effectiveness Evaluation

Developer-Ready Module & Feature Specification

Implementation handoff for Product, Workflow/UI, Backend, Frontend, QA, DevOps, and Codex-assisted development

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Audience	Product, Analyst, Backend, Frontend, QA, DevOps, Codex

Purpose: translate the approved business concept into one coherent development context that can be decomposed directly into epics, user stories, APIs, database entities, jobs, rules, and test cases.

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How to use this document

Treat requirement IDs, enums, state transitions, configuration rules, and acceptance criteria as the baseline. Technical implementation may vary by stack, but business behavior must remain consistent. No thresholds, role rules, or scoring weights should be hard-coded when this document marks them configurable.

1. Recommended Module Name and Positioning

1.1 Recommended name

Q-VOICE - Quilvian Patient Voice & Experience Intelligence

Q-VOICE is the recommended product name because the module captures the patient voice, converts it into measurable experience data, resolves complaints, and transforms recurring patterns into management recommendations and evaluated improvement actions.

1.2 Position in the Quilvian ecosystem

Layer	Q-VOICE role
Quilvian Patient Journey	Receives patient-journey milestones and attaches feedback to the correct encounter and service touchpoint.
Patient Engagement	Captures score, comments, complaint, attachment, callback consent, additional information, and resolution feedback.
Hospital Operations	Creates structured complaint cases, worklists, SLA timers, escalation, service recovery, and closure.
Management Intelligence	Aggregates complaint patterns, ranks issues, proposes recommendations, tracks decisions, and evaluates outcomes.
Quality & Patient Safety	Links safety concerns, RCA/CAPA, systemic issues, recurrence, and effectiveness review.
iHOS / Core Systems	Consumes patient, registration, encounter, unit, physician, payer, billing, pharmacy, and diagnostic context.

1.3 Feature group names

Feature group	Recommended label	Purpose
Patient-facing	My Experience	Score the visit, send feedback/complaint, track resolution.
Operational	Complaint & Service Recovery	Triage, assign, investigate, communicate, resolve, and close.
Analytics	Experience Intelligence	Scores, trends, heatmaps, sentiment, recurrence, and issue detection.
Management	Recommendation Center	Evidence-based management issues, recommendations, approval, and prioritization.
Improvement	Action & Effectiveness	Action plan, CAPA linkage, baseline-target monitoring, effectiveness review.

Feature group	Recommended label	Purpose
Administration	Q-VOICE Configuration	Master data, survey templates, thresholds, SLA, rules, roles, and integrations.

1.4 Alternative names

Name	Strength	Why not primary
Quilvian Patient Experience Hub	Very clear and familiar	Does not strongly communicate management intelligence.
Quilvian Voice of Patient	Strong healthcare terminology	Less explicit about complaint resolution and evaluation.
Quilvian Experience & Recovery	Strong service-recovery positioning	Less suitable for analytics and improvement governance.
Quilvian PX Intelligence	Short and modern	PX abbreviation may not be immediately understood by all hospital users.

2. Product Summary

Q-VOICE is a closed-loop patient experience and complaint intelligence module. It begins with patient feedback at a specific care touchpoint, converts negative signals into structured complaint cases, manages service recovery with SLA and escalation, then analyzes patterns to produce management issues and improvement recommendations. Approved actions are measured against baseline and target until the system can determine whether the intervention was effective.

Closed-loop outcome

Listen -> Measure -> Detect -> Triage -> Recover -> Investigate -> Resolve -> Aggregate -> Recommend -> Approve -> Improve -> Evaluate -> Standardize / Rework.

2.1 Core business outcomes

- One patient voice record linked to the correct patient journey and encounter.
- One structured complaint workflow with clear ownership, SLA, escalation, and audit trail.
- One management view that explains what is happening, how important it is, and what should be done.
- One improvement loop where a completed action is not considered successful until outcome evidence improves.
- One consistent data model that supports multi-hospital, multi-unit, and cross-period comparison.

2.2 Product boundaries

In scope	Out of scope / controlled boundary
Survey, score, complaint, service recovery, recommendation, action plan, effectiveness	Replacement of EMR, registration, billing, pharmacy, LIS, RIS, or incident reporting.
AI-assisted classification/recommendation when enabled	Autonomous clinical diagnosis, legal admission, staff sanction, compensation approval.
Operational and management dashboards	Public publication of patient complaint detail.
Safety flag and linkage to Quality process	Q-VOICE does not replace patient safety incident governance.

3. Scope and Design Principles

3.1 Design principles

Principle	Implementation meaning
Patient-first	Patient flow must be short, mobile-first, plain language, and independent of hospital org structure.
Closed-loop	Every material issue must have owner, due date, action, outcome measurement, and clear closure state.
Configuration-first	Categories, weights, SLA, thresholds, templates, routing, and evaluation periods must be configurable.
Human-in-control	AI can suggest but cannot make clinical, disciplinary, legal, compensation, or executive decisions.
Traceability	Every important decision must be attributable to user/rule/evidence and timestamp.
Separation of views	Internal notes and sensitive reviews must never leak into patient-facing communication.
Event-driven	Core state changes publish events for notification, analytics, integration, and audit.
Resilient integration	Patient feedback should remain submittable even when a source system is temporarily unavailable.

3.2 Supported care journeys

- Outpatient
- Emergency department
- Inpatient / daily pulse
- Surgery / procedure
- Pharmacy
- Laboratory
- Radiology
- MCU / corporate health
- Discharge / post-discharge
- Other configurable journey types

4. Module and Feature Map

ID	Menu Group	Feature	Purpose
QV-01	My Experience	Patient Survey	Capture 1-5 score, domain questions, NPS, comment, consent.
QV-02	Complaint Intake	Complaint Submission	Create complaint from patient, staff-assisted, or external channel.
QV-03	Triage	Classification & Severity	Validate, classify, risk flag, determine severity and routing.
QV-04	Case Workspace	Complaint Case Management	Owner, SLA, investigation, timeline, communication, evidence.
QV-05	Recovery	Service Recovery	Immediate recovery, resolution, compensation record, closure approval.
QV-06	Experience	Experience Scoring	Domain score, overall score, NPS, response rate, journey analytics.
QV-07	Issue Center	Management Issue Detection	Aggregate recurring complaints into systemic issues.
QV-08	Recommend	Recommendation Engine	Rank issues and propose action options with evidence.
QV-09	Decision	Management Decision	Approve, revise, defer, reject, budget, escalate.
QV-10	Improve	Action Plan / CAPA	Owner, due date, KPI, evidence, progress, blockers.
QV-11	Evaluate	Effectiveness Review	Compare baseline, target, current result, sustainability.
QV-12	Insights	Dashboard & Reports	Operational, unit, Quality, executive, management insight.
QV-13	Notify	Notification & Escalation	Patient updates, owner alerts, SLA warnings, critical escalation.
QV-14	Integrate	Integration & Sync	iHOS context, messaging, SSO, Quality, DWH.
QV-15	Govern	Configuration & Audit	RBAC, master data, thresholds, templates, audit, versioning.

5. Actors and Access Model

Actor	Primary capability	Restrictions
Patient / Family	Survey, complaint, evidence, tracking, resolution feedback	Only own case; internal notes hidden.
Frontliner	Assisted complaint intake and immediate recovery	Limited case detail; cannot close high-risk case.
Customer Care	Validate, triage, assign, communicate, resolve	No clinical/legal decision beyond authority.
Unit PIC	Investigate assigned case, actions, evidence	Only assigned unit/case unless additional permission.
Unit Head / Manager	Approve unit resolution, action plan, monitor KPI	Unit scope by default.
Quality / Patient Safety	Safety review, RCA/CAPA, recurrence, effectiveness	Cross-unit quality access.
Medical / Nursing Governance	Professional review	Restricted cases only.
Finance / Insurance	Billing/payer investigation and financial recovery	Financial scope only.
HR / Legal / Marcomm	Specialized restricted-case handling	Need-to-know access.
Directors	Enterprise insight, critical case, recommendation decision	Full oversight subject to sensitive-case rules.
System Admin	Configuration, integration health, user-role support	No business override by default.
Auditor	Read-only review of authorized records and audit trail	No mutation.

5.1 Permission domains

- patient_feedback.view_own / create
- complaint.create / view / assign / classify / investigate / resolve / close / reopen
- complaint.sensitive.view
- recommendation.view / generate / decide
- action_plan.create / update / approve / evaluate
- dashboard.unit / enterprise
- configuration.manage
- export.execute
- audit.view

6. End-to-End Business Flow

#	Stage	System behavior	Output
1	Encounter milestone	iHOS/Patient Journey emits eligible event or staff opens assisted form.	Survey eligibility / complaint context
2	Patient feedback	Patient submits score, comment, complaint, attachment, and contact consent.	Feedback record
3	Low-score trigger	Score/risk rule evaluates response.	Complaint offer or auto-ticket
4	Case validation	Customer Care validates identity, encounter, duplicate, chronology, expectation.	Validated case
5	Triage	Category, subcategory, severity, risks, sentiment, owner, SLA.	Assigned case
6	Immediate recovery	Patient receives acknowledgement and urgent recovery where possible.	Recovery record
7	Investigation	Unit/PIC records evidence, findings, causes, and impact.	Investigation record
8	Resolution	Resolution is approved as required and communicated to patient.	Resolution / closure
9	Issue detection	Analytics groups recurring/similar complaints.	Management issue
10	Recommendation	System builds evidence and recommendation options.	Recommendation package
11	Decision & action	Management decides and approved option becomes action plan.	Action plan
12	Evaluation	Baseline vs target/current measured at scheduled checkpoints.	Effectiveness result
13	Learning loop	Effective action standardized; ineffective action revised/escalated.	Knowledge / revised control

7. Patient-Side Functional Specification

7.1 Entry points

Req ID	Function	Behavior	Priority
PAT-001	QR / web survey	Secure token opens survey without mandatory account login.	Must
PAT-002	Patient app / portal	Authenticated user sees eligible recent encounters.	Should
PAT-003	Message invitation	WhatsApp/SMS/app notification can deep-link to one encounter.	Should
PAT-004	Assisted complaint	Staff can enter phone/desk/email/paper complaint and source is retained.	Must

7.2 Survey screen behavior

Req ID	Requirement	Acceptance behavior
PAT-010	Overall score first	Show 1-5 with text labels. Patient can continue without rating non-applicable domains.
PAT-011	Dynamic domain questions	Question set determined by journey, unit, encounter, and active survey template version.
PAT-012	N/A handling	N/A excluded from score denominator and stored explicitly.
PAT-013	Low score branch	Configured low score opens improvement category + narrative + optional complaint flow.
PAT-014	Positive branch	High score allows compliment and optional testimonial/recognition consent.
PAT-015	Free text	Allow comment with character limit, language-safe storage, and abuse-safe input validation.
PAT-016	Duplicate control	Prevent or mark repeated response according to token/encounter policy.
PAT-017	Consent	Display privacy notice and separate contact/testimonial consent where needed.
PAT-018	Confirmation	Display reference ID, submission time, and next-step expectation after submit.

7.3 Complaint form fields

Section	Required	Optional / conditional
Reporter	Anonymous flag or reporter name; relationship; preferred contact when follow-up requested	Email, alternate phone, preferred time
Patient / Encounter	Encounter token or MRN/registration match when available	Manual encounter detail when source unavailable
Incident	Incident date/time or approximate period; unit/location; primary category	Staff name/description, subcategory
Narrative	What happened; impact; desired outcome	Attachments/photos/documents
Consent	Privacy acknowledgement; permission to contact if requested	Permission to use anonymized feedback

7.4 Patient tracking

Internal status	Patient-visible status	Patient capability
NEW / AWAITING_VALIDATION	Laporan diterima	View reference and receipt time.
ASSIGNED / UNDER_INVESTIGATION	Sedang ditindaklanjuti	View next expected update date.
WAITING_PATIENT	Membutuhkan informasi tambahan	Submit text/file response.
ACTION_IN_PROGRESS	Perbaikan sedang dilakukan	Receive progress update where appropriate.
RESOLUTION_COMMUNICATED	Hasil tindak lanjut tersedia	Read resolution and provide satisfaction score.
CLOSED	Selesai	View summary; request reopen within configured period.
REOPENED	Ditinjau kembali	View new update cycle.

7.5 Patient UX rules

- Mobile-first; standard survey target under 3 minutes.
- No internal severity, legal flag, staff disciplinary note, or management recommendation shown to patient.
- Status labels use plain Indonesian, not workflow jargon.
- Critical clinical text must not be placed in unsecured notification preview.
- Attachment upload must show progress and failure recovery.
- Patient can submit complaint even if the encounter cannot immediately be matched; reconciliation happens later.

8. Management-Side Functional Specification

8.1 Work queue

Req ID	Requirement	Minimum columns / behavior
MGT-001	Role-based queue	Complaint ID, patient, unit, category, severity, status, owner, SLA remaining, last update.
MGT-002	Filtering	Site, unit, journey, category, severity, status, owner, date, payer, score, risk.
MGT-003	Priority ordering	Critical first, overdue, near SLA, callback required, then configured business priority.
MGT-004	Saved views	User/role may save filters and column layout.
MGT-005	Bulk actions	Only permitted low-risk operations; each mutation audited.

8.2 Complaint case workspace

Area	Required content
Header	Complaint ID, severity badge, status, patient/encounter, unit, owner, SLA timer, risk flags.
Overview	Summary, category, patient expectation, source, score, journey context.
Timeline	Immutable chronological activity: status, assignment, communication, evidence, decisions.
Communication	Inbound/outbound log with patient-visible content separate from internal notes.
Investigation	Evidence sources, interviews, records reviewed, findings, cause, impact, limitation.
Service Recovery	Immediate action, resolution, compensation/waiver if any, authority/approval.
Linked Records	Survey, complaints, systemic issue, incident/RCA, action plan, evaluation.
Audit	Who viewed/exported/changed critical fields and before/after values.

8.3 Management decision center

Req ID	Requirement	Behavior
MGT-050	Issue evidence card	Volume, rate, trend, severity, score movement, units, shifts, recurrence, representative anonymized comments.
MGT-051	Priority score	Configurable weighted score with visible component values.
MGT-052	Recommendation options	One or more action options with rationale, impact, effort, cost band, owner, target.
MGT-053	Decision workflow	Approve, approve with revision, request analysis, defer, reject, budget plan, escalate.
MGT-054	Decision audit	Decision reason, approver, timestamp, conditions, previous recommendation version.
MGT-055	Action conversion	Approved recommendation becomes an action plan without retyping evidence and target.

9. Complaint State Machine and SLA

9.1 Internal states

State	Entry meaning	Allowed next step
NEW	Ticket created	Start validation
AWAITING_VALIDATION	Waiting triage	Validate / duplicate / cancel
VALIDATED	Business-valid complaint	Assign owner
ASSIGNED	Primary owner exists	Owner acknowledges
UNDER_INVESTIGATION	Investigation active	Complete findings / request info
WAITING_PATIENT	Patient input needed	Patient responds / timeout
WAITING_DECISION	Management/committee input required	Decision captured
ACTION_IN_PROGRESS	Recovery/corrective action underway	Action complete
RESOLUTION_PROPOSED	Draft resolution prepared	Approval / communicate
RESOLUTION_COMMUNICATED	Patient informed	Feedback window
WAITING_CONFIRMATION	Awaiting patient response	Close / reopen
CLOSED	Closure checklist satisfied	Reopen only
REOPENED	New evidence/disagreement	Return to investigation
CANCELLED_DUPLICATE	Invalid or duplicate	Terminal

9.2 Severity and default SLA

Level	Typical examples	Initial response	Investigation / interim	Resolution
L1 LOW	Facility, minor information	<= 4 hours	As needed	<= 3 business days
L2 MODERATE	Courtesy, delay, admin	<= 2 hours	Plan <= 1 business day	<= 2 business days
L3 HIGH	Material billing, repeated failure, potential safety	<= 30 minutes	Initial <= 4 hours	Interim/final <= 24 hours
L4 CRITICAL	Serious harm, sentinel, violence, privacy, legal/media	Real-time; management <= 15 min	Team <= 1 hour; interim report <= 6 hours	Executive case plan

9.3 SLA clock rules

- Business-hours versus 24x7 calculation must be configurable by severity/category/site.
- Near-due thresholds default to 75% and 90% SLA consumption and are configurable.
- SLA pause requires allowed pause reason, authorizing role, timestamp, and resume event.
- Changing severity recalculates remaining target according to configured policy; historical target is retained.
- Critical escalation is fan-out to all required roles, not sequential routing.

10. Patient Experience Scoring

10.1 Default domains

Domain	Examples	Default weight
Access	Booking, registration, contact, navigation	8%
Waiting Time	Doctor, diagnostic, pharmacy, discharge	12%
Doctor Experience	Listening, explanation, shared decision	15%
Nursing Experience	Response, explanation, pain, safety	15%
Staff Courtesy	Registration, cashier, security, pharmacy	8%
Communication	Clarity, consistency, next steps	12%
Facility	Cleanliness, comfort, toilet, parking	8%
Safety & Trust	Identity, privacy, confidence	10%
Billing & Administration	Transparency, payment, payer process	6%
Discharge	Medication, warning signs, follow-up	6%

Domain scoring

Domain Score = $\text{sum}(\text{valid response values}) / (\text{valid response count} \times \text{maximum scale}) \times 100$. N/A responses are stored but excluded from the denominator.

Overall scoring

Overall Experience Score = $\text{sum}(\text{Domain Score} \times \text{active configured domain weight})$. Weight set must be versioned by effective date.

10.2 Score safeguards

- Do not silently mix different scale versions or survey versions in one unqualified trend.
- Scores are analytics, not substitutes for critical-event governance. A high average score cannot suppress L3/L4 escalation.
- Store raw answers separately from derived scores to allow recalculation after approved formula changes.
- Filterable denominator must be visible in dashboards to avoid misleading small-sample comparisons.

11. Management Issue and Recommendation Engine

11.1 Management issue detection

Management Issue

A management issue is a persistent or material pattern created from one or more complaint/experience signals that warrants management attention beyond individual case closure.

Detection signal	Example rule
Frequency	Same primary category + unit exceeds configured count/rate in window.
Recurrence	Similar complaint reappears after related action was implemented.
Severity concentration	Multiple L3 or any qualifying L4 in defined period.
Score decline	Domain or unit score drops by configured points/percentage.
Cross-unit pattern	Same process failure appears across multiple units.
Operational correlation	Complaint aligns with high wait time, downtime, staffing, TAT, or billing metric.
Manual creation	Authorized manager/Quality creates issue from evidence not detectable automatically.

11.2 Default priority score

Formula

Priority Score = Frequency 25% + Severity 25% + Patient Impact 20% + Repeat Rate 15% + Legal/Reputation Risk 15%. Each component is scored 1-5. Weights and bands are configuration, not code constants.

Score	Band	Default action
4.25-5.00	CRITICAL	Immediate executive decision.
3.50-4.24	HIGH	Action plan within 3 business days.
2.50-3.49	MEDIUM	Unit improvement within 14 days.
<2.50	LOW	Monitor or routine improvement.

11.3 Recommendation output

Field	Required behavior
Issue statement	Short problem, scope, period, affected journeys/units.
Evidence	Counts, rates, trends, score change, severity, operational data, anonymized comments.
Probable cause	Rule/analytic/AI suggestion with source and confidence; not final RCA unless approved.
Options	At least one actionable option; alternatives allowed.
Impact / effort	High/medium/low with reason.
Cost band	None/low/medium/high or hospital-configured range.
Owner / sponsor	Proposed responsible role and management sponsor.
Target	Baseline, target KPI, source, formula, due date.
Decision	Approve / revise / analyze / defer / reject / budget / escalate.
Version	Recommendation revision number and decision linkage.

11.4 AI guardrails

- Always label AI result as a recommendation requiring human review.
- No AI-generated sanction, legal admission, compensation approval, or clinical conclusion can be auto-executed.
- AI-derived category/cause/recommendation must store model/rule version, confidence, input references, and human override.
- Sensitive patient identifiers must remain inside approved environment; no unapproved external training or inference path.

12. Action Plan and Effectiveness Evaluation

12.1 Action plan fields

Field group	Required fields
Identity	action_plan_id, management_issue_id, recommendation_id/version
Accountability	owner, co-owner, sponsor, approver
Plan	action title, description, start, due date, cost/budget, dependencies
Measurement	baseline, target, KPI formula, data source, review schedule
Progress	status, percent, blocker, evidence, updated_at
Outcome	current value, effectiveness score, conclusion, next action

12.2 Action status

- PROPOSED
- APPROVED
- NOT_STARTED
- IN_PROGRESS
- AT_RISK
- DELAYED
- IMPLEMENTED
- UNDER_EVALUATION
- EFFECTIVE
- ADEQUATE
- PARTIALLY_EFFECTIVE
- INEFFECTIVE
- CLOSED

12.3 Default effectiveness formula

Formula

Effectiveness Score = Complaint Reduction 30% + Experience Improvement 25% + KPI Achievement 25% + Sustainability 20%. Components, weights, and thresholds are configurable and versioned.

Result	Default range	System response
EFFECTIVE	>=85	Offer standardization / replication; continue recurrence monitoring.

Result	Default range	System response
ADEQUATE	70-84	Continue with refinements and next review.
PARTIALLY_EFFECTIVE	50-69	Require revised action or targeted follow-up.
INEFFECTIVE	<50	Require RCA/recommendation review and management escalation.

12.4 Evaluation schedule

Checkpoint	Purpose
Day 7	Implementation started; blockers and process adherence.
Day 14	Early process compliance and staff adoption.
Day 30	Initial outcome impact.
Day 60	Stability and trend.
Day 90	Sustainability / standardization decision.
Month 6	Long-term recurrence and replication review.

13. Information Architecture and Key Screens

13.1 Main management navigation

Menu	Purpose
Dashboard	Operational and executive overview.
Complaints	Queue, case search, case workspace, SLA.
Experience	Survey results, score trends, comments, response rate.
Issues	Systemic management issues and priority matrix.
Recommendations	Recommendation packages and decision queue.
Action Plans	Approved improvements, CAPA linkage, progress.
Evaluation	Due reviews, effectiveness results, recurrence.
Reports	Operational, quality, executive, audit, export.
Administration	Survey templates, master data, SLA, thresholds, roles, integrations, audit.

13.2 Patient screens

Screen ID	Screen	Key elements
P01	Survey Landing	Encounter context, privacy note, start CTA.
P02	Overall Experience	1-5 score with label, optional comment.
P03	Domain Questions	Dynamic question groups and N/A.
P04	Low Score / Complaint	Category, narrative, impact, expected outcome, contact consent.
P05	Attachment	File list, progress, validation, remove.
P06	Confirmation	Reference ID, next expected update, track CTA.
P07	Track Complaint	Friendly status, last update, expected next update.
P08	Provide Information	Reply text/file to WAITING_PATIENT request.
P09	Resolution Feedback	Resolution summary, handling score, reopen request.

13.3 Management screens

Screen ID	Screen	Key elements
M01	Complaint Queue	Filters, severity, SLA, owner, callback, saved views.
M02	Complaint Detail	Header, overview, timeline, communication, investigation, recovery, linked records.
M03	Issue Center	Issue cards, evidence, priority, linked cases, trend.
M04	Recommendation Detail	Evidence, cause hypothesis, options, impact-effort, cost, target, decision.
M05	Action Plan	Plan, owners, KPI, timeline, evidence, blockers.
M06	Evaluation Review	Baseline, target, actual, component score, conclusion, next action.
M07	Dashboard	KPI, heatmap, priority issues, overdue, effectiveness.
M08	Configuration	Templates, categories, SLA, weights, rules, roles, integrations.

14. Data Model and Core Entities

14.1 Core entities

Entity	Minimum fields
patient_experience_response	survey_id, patient_id, encounter_id, template_version, journey, channel, submitted_at, overall_score, nps, comment, consent
experience_answer	answer_id, survey_id, question_id, domain_id, value, not_applicable, raw_text
complaint	complaint_id, survey_id, patient_id, encounter_id, reporter, incident_at, source, category_id, severity, risks, owner, state, sla fields
complaint_category_link	complaint_id, category_id, subcategory_id, primary_flag
complaint_investigation	complaint_id, investigator, sources, findings, cause, impact, completed_at
complaint_communication	complaint_id, direction, channel, patient_visible, content_summary, sent_at, delivery_status
complaint_attachment	attachment_id, complaint_id, type, storage_ref, classification, hash, uploader, uploaded_at
complaint_action	action_id, complaint_id, action_type, owner, due_at, status, evidence
management_issue	issue_id, title, scope, detection_rule, priority_score, band, status
management_issue_case	issue_id, complaint_id, contribution_weight/reason
recommendation	recommendation_id, issue_id, version, rationale, option, impact, effort, cost_band, proposed_owner, target
management_decision	decision_id, recommendation_id/version, decision, reason, approver, conditions, decided_at
action_plan	action_plan_id, issue_id, recommendation_id, owner, sponsor, baseline, target, start/due, status
evaluation	evaluation_id, action_plan_id, checkpoint, component_results, effectiveness_score, conclusion, next_action
audit_event	audit_id, actor, action, entity, entity_id, before, after, source, trace_id, occurred_at

14.2 Entity relationship summary

```

Patient -> Encounter -> Survey -> Answers
    |
    +-> Complaint -> Investigation / Communication / Action / Attachments
        |
        +-> Management Issue <- other Complaints
            |
            +-> Recommendation(versioned) -> Decision
                |
                +-> Action Plan -> Evaluation(s)
    
```

14.3 Key enums

Enum	Values / notes
Severity	LOW, MODERATE, HIGH, CRITICAL
ComplaintState	NEW, AWAITING_VALIDATION, VALIDATED, ASSIGNED, UNDER_INVESTIGATION, WAITING_PATIENT, WAITING_DECISION, ACTION_IN_PROGRESS, RESOLUTION_PROPOSED, RESOLUTION_COMMUNICATED, WAITING_CONFIRMATION, CLOSED, REOPENED, CANCELLED_DUPLICATE
RiskType	CLINICAL_SAFETY, LEGAL, PRIVACY, REPUTATION, MEDIA, FINANCIAL, VIOLENCE, OTHER
Decision	APPROVE, APPROVE_WITH_REVISION, REQUEST_ANALYSIS, DEFER, REJECT, BUDGET_PLAN, ESCALATE
Effectiveness	EFFECTIVE, ADEQUATE, PARTIALLY_EFFECTIVE, INEFFECTIVE
Channel	WEB, QR, APP, WHATSAPP, SMS, PHONE, DESK, EMAIL, PAPER, SOCIAL_REVIEW, OTHER

15. Business Rules and Configuration

Rule	Requirement
BR-001	Overall or configured domain score 1-2 offers complaint and callback workflow.
BR-002	Safety-related low score or critical keyword creates a review flag; final severity requires validated rule/human review.
BR-003	Complaint can have multiple categories but exactly one primary category and one primary owner.
BR-004	CRITICAL case cannot be closed solely by the reported unit.
BR-005	Closure requires resolution note, patient communication record, and severity-specific approvals.
BR-006	HIGH/CRITICAL require investigation evidence; CRITICAL requires configured executive/Quality approval.
BR-007	Severity downgrade requires reason; original severity is immutable in audit history.
BR-008	Configured recurrence rule may create/update a management issue; default starter rule may be three similar complaints in 30 days.
BR-009	Complaint communication may close while linked CAPA/action plan remains open.
BR-010	Anonymous complaint may proceed, but contact-dependent steps are N/A.
BR-011	Recommendation remains advisory until management decision.
BR-012	Action plan cannot become EFFECTIVE merely because tasks are completed; outcome evidence is mandatory.
BR-013	INEFFECTIVE action requires re-analysis, revised action, or explicit accepted-risk decision.
BR-014	Patient-visible text cannot expose internal disciplinary, peer review, legal strategy, or confidential investigation detail.
BR-015	Inactive master values remain resolvable for historical records.
BR-016	All scoring weights, thresholds, SLA, category routing, and templates are effective-dated and versioned.
BR-017	System must distinguish complaint count, complaint item count, patient count, and encounter count.
BR-018	No hard delete of complaint, recommendation, decision, evaluation, or audit record.

15.1 Configurable master data

- Hospital/site, unit, service journey, physician/staff reference, payer
- Survey template, domain, question, scale, weight set
- Complaint category/subcategory, severity matrix, risk types
- SLA calendar, pause reasons, escalation matrix, routing rules
- Communication templates, channel availability, languages
- Recommendation taxonomy, priority weights, cost bands

- KPI definitions, evaluation formula, checkpoints, effectiveness bands
- Roles, access scopes, restricted-case groups

16. Event Model, Jobs, and Notifications

16.1 Domain events

Event	Publisher	Main consumers
encounter.completed	iHOS / Patient Journey	Survey eligibility service
survey.eligible	Q-VOICE	Invitation scheduler
survey.submitted	Q-VOICE	Scoring, complaint trigger, analytics
complaint.created	Q-VOICE	Triage queue, notification, audit
complaint.classified	Q-VOICE	Routing, SLA, analytics
complaint.assigned	Q-VOICE	Owner notification
complaint.severity_changed	Q-VOICE	Escalation, audit
complaint.sla_warning	Scheduler	Owner/supervisor notification
complaint.resolution_communicated	Q-VOICE	Patient feedback window
complaint.closed	Q-VOICE	Issue detection, analytics
issue.created_or_updated	Analytics	Recommendation engine
recommendation.generated	Recommendation Engine	Decision queue
recommendation.approved	Management	Action plan service
action.evaluation_due	Scheduler	Owner/Quality
action.evaluated	Evaluation	Dashboard, learning loop

16.2 Background jobs

Job	Frequency / trigger	Behavior
Survey eligibility	Event-driven + reconciliation	Create eligible survey with expiry/token.
Invitation sender	Scheduled queue	Send template; retry provider-transient errors.
SLA monitor	At least every few minutes or event-driven timer	Publish near-due/overdue events.
Encounter matcher	On submit + retry queue	Resolve unmatched complaint to patient/encounter.
Issue detector	Near-real-time or scheduled batch	Evaluate recurrence/trend rules.
Evaluation scheduler	Daily + action events	Create due review tasks and auto-pull metrics when available.
Integration reconciliation	Scheduled	Retry failed sync, detect stuck transactions, avoid duplicates.

16.3 Notification triggers

Trigger	Recipient	Minimum payload
Survey invitation	Patient	Service context, secure link, expiry.
Complaint received	Reporter	Reference, receipt time, next update.
Case assigned	Owner	Severity, due time, summary, case link.
Critical case	Configured response group	Immediate action, site/unit, safe summary, case link.
Near SLA / overdue	Owner / supervisor / manager	Remaining/breached time, blocker, case link.
Information requested	Patient	Requested item, safe instruction, due date.
Resolution available	Patient	Resolution summary link and feedback CTA.
Evaluation due	Action owner / Quality	Baseline, target, checkpoint, due date.

17. API and Integration Contract

17.1 API principles

- REST/JSON or platform-standard contract; exact protocol follows hospital architecture.
- All mutations support idempotency key or equivalent deduplication.
- Every response/error includes correlation/trace ID.
- Authorization enforced server-side; UI visibility is not security.
- Do not expose internal notes through patient endpoints.
- Version public/integration APIs; breaking change requires controlled migration.

17.2 Suggested service endpoints

Method	Endpoint	Purpose
POST	/api/v1/surveys/{token}/responses	Submit patient survey.
GET	/api/v1/patient/complaints/{reference}	Patient-safe complaint status.
POST	/api/v1/patient/complaints/{reference}/responses	Provide requested patient info.
POST	/api/v1/complaints	Create staff-assisted complaint.
GET	/api/v1/complaints	Management queue/search.
GET	/api/v1/complaints/{id}	Internal complaint detail.
POST	/api/v1/complaints/{id}/classify	Classify/severity/risk.
POST	/api/v1/complaints/{id}/assign	Assign owner.
POST	/api/v1/complaints/{id}/transition	Controlled state transition.
POST	/api/v1/complaints/{id}/communications	Record/send communication.
POST	/api/v1/complaints/{id}/investigation	Save investigation/findings.
POST	/api/v1/issues	Manual management issue create.
GET	/api/v1/issues	Issue center search.
POST	/api/v1/issues/{id}/recommendations/generate	Generate/recompute advisory recommendation.
POST	/api/v1/recommendations/{id}/decisions	Record management decision.
POST	/api/v1/action-plans	Create action plan.
POST	/api/v1/action-plans/{id}/evaluations	Record effectiveness review.
GET	/api/v1/config/*	Read configuration.
POST	/api/v1/config/*	Versioned config mutation by authorized admin.

17.3 Example complaint creation payload

```
POST /api/v1/complaints
Idempotency-Key: <uuid>
{
  "source": "WEB",
```

```

"survey_id": "SVY-...",
"patient_id": "PAT-...",
"encounter_id": "ENC-...",
"reporter": {"relationship": "SELF", "contact_consent": true},
"incident_at": "2026-08-12T10:30:00+07:00",
"category_ids": [{"id": "WAITING_TIME", "primary": true}],
"narrative": "...",
"expected_outcome": "..."
}
    
```

17.4 iHOS integration minimum

Integration	Direction	Minimum data
Patient Master	Inbound	patient_id/MRN, name, DOB, contact subject to access.
Encounter / Registration	Inbound	encounter_id, service date, journey, site, unit, physician, room, payer, status.
Billing / Pharmacy / Diagnostic context	Inbound optional	Operational data used as evidence, not blindly copied into patient messages.
SSO / Identity	Bi-directional	User identity, role/group mapping, deactivation.
Messaging Gateway	Bi-directional	Send, delivery status, inbound reply if supported.
Quality / Incident System	Bi-directional	Safety flag, incident/RCA/CAPA link and status.
DWH / BI	Outbound	Approved analytics dataset, preferably de-identified for broad use.

18. Security, Privacy, and Audit

Control	Minimum requirement
RBAC	Least privilege; site/unit/case sensitivity scope evaluated server-side.
Sensitive cases	Restricted visibility for violence, harassment, legal, privacy, staff discipline, other configured categories.
Encryption	Transport and storage encryption using hospital-approved standards.
Attachment security	Virus/malware scanning, type/size allowlist, secure object reference, no public URL.
Audit	Log view of restricted case, create/edit, severity change, assignment, decision, export, config change, failed access.
Patient endpoints	Use secure token/authentication and patient-safe projection only.
Exports	Permission-gated, logged, optionally watermarked with user/time.
Retention	Configurable retention, archive, legal hold; no hard delete for regulated business records.
Secrets	No API secrets, tokens, or provider keys in client code or logs.
AI privacy	No sensitive data leaves approved environment without explicit architecture/security approval.

18.1 Audit event minimum

```
{
  "actor_id": "USR-...",
  "action": "COMPLAINT_SEVERITY_CHANGED",
  "entity_id": "CMP-...",
  "before": {"severity": "MODERATE"},
  "after": {"severity": "HIGH"},
  "reason": "Safety impact identified",
  "trace_id": "...",
  "occurred_at": "..."
}
```

19. Error Handling and Resilience

Condition	Required behavior
iHOS unavailable during patient submit	Accept feedback/complaint with available context; queue encounter reconciliation.
Duplicate submit / network retry	Idempotency returns original result; do not create duplicate complaint.
Messaging provider failure	Retry transient error; mark permanent failure; create fallback task when required.
Attachment upload interrupted	Resume/retry or clear error; complaint may save without optional attachment.
Rules engine unavailable	Do not silently classify as low risk; place in manual triage state and alert operations.
Recommendation engine unavailable	Complaint operations continue; issue waits for manual/system recommendation.
AI low confidence	Require human classification; do not auto-overwrite human decision.
SLA scheduler delayed	Recalculate based on absolute timestamps after recovery; never reset clock.
Integration duplicate event	Use source event ID/idempotency key and ignore duplicate side effects.
Conflicting master versions	Use effective-dated version tied to transaction; raise config integrity alert.

20. Non-Functional Requirements

NFR	Target / requirement
Availability	Target >=99.5% monthly excluding approved maintenance; final production SLA agreed with vendor.
Patient submit reliability	No loss of acknowledged submission.
Common screen performance	P95 <=3 seconds under agreed production load for normal queries.
Mutation response	P95 <=5 seconds excluding file upload/provider latency.
Scalability	Multi-site, multi-unit, multi-journey; capacity sized for at least 3-year projected volume.
Configurability	New category, threshold, SLA, weight, template, routing rule without application redeploy.
Observability	Structured logs, metrics, trace IDs, integration health, queue depth, job lag, failure rate.
Recovery	Backup/restore and RPO/RTO proposed by vendor and proven before go-live.
Localization	Indonesian UI; architecture supports additional languages; timestamps shown in hospital timezone.
Accessibility	Readable contrast, labels, keyboard-friendly staff portal, accessible patient forms.
Data portability	Documented API/export and hospital-owned data; avoid proprietary lock-in.
Testability	Business rules callable/testable independently; time and configuration injectable in tests.

21. Analytics, Dashboard, and Reporting

21.1 Executive KPIs

- Overall Experience Score and domain scores
- NPS and response rate
- Complaint rate per 1,000 encounters
- Open, overdue, L3/L4 complaints
- First response SLA and resolution SLA
- Top categories, units, journeys, shifts
- Repeat complaint and reopen rate
- Resolution satisfaction
- Priority management issues
- Action plan at risk / overdue
- Effectiveness result and recurrence after action

21.2 Operational dashboards

Dashboard	Minimum widgets
Customer Care	New, unassigned, callback queue, near-due, overdue, waiting patient, reopened, workload.
Unit Head	Unit score, latest comments, category trend, active cases, root causes, actions, staff/shift pattern.
Quality	Safety flags, L3/L4, RCA/CAPA links, recurrence, systemic issues, ineffective actions.
Management Recommendation	Priority matrix, impact-effort, decision pending, cost band, owner, target.
System Admin	API health, failed sync, queue depth, oldest job, notification failures, config/audit changes.

21.3 Analytics rules

- Always expose denominator/sample size with percentages and scores.
- Distinguish complaint, complaint item, patient, encounter, and management issue counts.
- Trend comparisons must respect survey/config version or show the version boundary.
- Restricted/sensitive dimensions must be masked or suppressed for users without access.
- Exports are permission-gated and audited.

22. Acceptance Criteria and UAT

ID	Scenario	Expected result
UAT-001	Patient submits score 2 after outpatient encounter	Survey stored; low-score flow offered; complaint can be created and linked; acknowledgement/reference shown.
UAT-002	Patient submits complaint while iHOS is unavailable	Complaint accepted with temporary context; reconciliation later links encounter without duplicate.
UAT-003	Safety keyword / answer triggers review	Risk flag visible; configured alert is created; human review required.
UAT-004	Customer Care changes severity	Reason required; before/after audited; SLA/escalation recalculated according to rule.
UAT-005	Unit attempts to close CRITICAL case alone	Closure blocked; required approver displayed.
UAT-006	Patient sends requested information	Response attaches to case timeline and state can return to investigation.
UAT-007	Same issue repeats after previous improvement	System links cases to an existing/new management issue according to recurrence rule.
UAT-008	Recommendation approved	Decision record created; action plan is generated with copied evidence, baseline, target, owner, review dates.
UAT-009	Action completed but outcome misses target	System does not mark Effective; evaluation results in partial/ineffective and triggers review.
UAT-010	Patient reopens within allowed period	Original closure remains in history; state changes to REOPENED and new workflow cycle begins.
UAT-011	Unauthorized user opens sensitive case	Access denied and audit event recorded.
UAT-012	Duplicate API retry	Only one business record/side effect exists; retry returns original or safe idempotent result.
UAT-013	Dashboard score after config version change	Trend indicates version boundary and does not silently recalculate historical raw response using new weight unless selected.
UAT-014	AI recommendation overridden by manager	Human decision is authoritative; AI version/confidence and override reason remain traceable.

23. Development Breakdown for Codex

23.1 Recommended epic sequence

Epic	Scope	Suggested order
E1 - Foundation	Identity, RBAC, master data, audit, config versioning	1
E2 - Patient Survey	Eligibility, token, questions, scoring, submit	2
E3 - Complaint Intake	Create, attachments, encounter match, acknowledgement	3
E4 - Complaint Workflow	State machine, assignment, classification, SLA, timeline	4
E5 - Communication & Recovery	Patient-safe messages, service recovery, closure	5
E6 - Experience Analytics	Domain scores, NPS, response rate, filters	6
E7 - Management Issues	Recurrence/trend detection and issue center	7
E8 - Recommendation & Decision	Priority score, options, management approval	8
E9 - Action & Evaluation	Action plan, KPI, checkpoints, effectiveness	9
E10 - Dashboard & Reports	Role dashboards, exports, monitoring	10
E11 - Integrations	iHOS, messaging, Quality, DWH	Parallel after contracts
E12 - Hardening	Performance, security, recovery, observability, UAT	Final

23.2 Codex implementation guardrails

- Implement the complaint state machine in a domain/service layer. Do not allow arbitrary state writes from controllers or UI.
- Centralize SLA calculation in one tested service with injectable clock/calendar.
- Configuration is effective-dated/versioned; business thresholds are never scattered as constants.
- Create audit event inside the same transaction or reliable outbox flow as critical business mutation.
- Use idempotency for all externally retried creates and transitions.
- Patient API returns a dedicated safe DTO/projection. Never serialize the internal complaint entity directly.
- Recommendation and AI services return advisory output; decision endpoint is a separate authorization boundary.
- Store raw survey answers and derived metrics separately.
- Use domain events/outbox for notification and integration, not synchronous provider calls inside core transaction where avoidable.
- Every new endpoint must define permission, validation, error model, audit behavior, and idempotency behavior.

23.3 Suggested repository/domain boundaries

```

qvoice/
survey/      # eligibility, templates, answers, scoring
complaint/    # intake, state machine, SLA, assignment, investigation
communication/ # templates, patient-safe messages, delivery
experience/    # score aggregation, analytics
issue/        # recurrence/systemic issue detection
recommendation/ # priority, recommendation, management decision
improvement/   # action plan, KPI, evaluation
integration/    # ihos, messaging, quality, dwh adapters
governance/    # config versions, RBAC policies, audit
shared/        # identifiers, clock, result/error, events
    
```

23.4 Definition for each Codex task

Task input	Must include
Context	Relevant requirement IDs and business rule IDs.
Data	Entity fields, enum values, nullability, indexes/uniqueness assumptions.
API	Request/response, validation, authorization, idempotency, error codes.
Behavior	Allowed state transitions, events, audit record, notification side effects.
Tests	Happy path, permission failure, invalid state, duplicate retry, integration failure.
Observability	Trace ID, structured log, metric/error counter where applicable.

24. Open Decisions and Assumptions

ID	Decision / assumption	Owner
OD-001	Final deployment architecture and runtime stack.	Hospital IT / Vendor
OD-002	Exact iHOS APIs, event availability, and source-of-truth mapping.	Hospital IT / iHOS
OD-003	Messaging provider and supported inbound reply model.	IT / Operations
OD-004	Final severity matrix and SLA calendar by service/site.	Patient Experience / Quality
OD-005	Service recovery / compensation authority limits.	Management / Finance / Legal
OD-006	AI/NLP capabilities enabled in MVP versus later release.	Management / IT / Quality
OD-007	Retention, archive, legal hold, and export policy.	Legal / Privacy / IT
OD-008	Final score weights, priority weights, and effectiveness thresholds.	Management / Quality
OD-009	Historical complaint migration depth.	Business / IT
OD-010	Multi-hospital tenancy and benchmarking visibility.	Management / IT
OD-011	Whether low score creates complaint automatically or only offers complaint form by journey/category.	Patient Experience
OD-012	Patient reopening window and timeout behavior.	Patient Experience / Legal

25. Definition of Done

Release acceptance principle

A feature is not Done when only the screen works. It is Done when business state, data, permission, audit, integration side effects, failure handling, and test coverage match this specification.

- UI/UX approved and mapped to requirement IDs.
- Database migration and indexes reviewed.
- API contract documented and versioned.
- Authorization and sensitive-case rules tested server-side.
- State transition, SLA, and scoring unit tests pass.
- Audit events verified for critical mutations and exports.
- Idempotency and integration retry behavior tested.
- Patient-safe projections reviewed for privacy leakage.
- Dashboard denominator and filter logic validated against test dataset.
- UAT scenarios applicable to the feature pass.
- Monitoring/logging/alerts are present for production support.
- Configuration and operating guide updated.

25.1 Recommended first development slice

1. Foundation: identity/RBAC, configuration versioning, audit/outbox.
2. Patient survey: eligibility, survey token, dynamic questions, raw answers, scoring.
3. Complaint: create/link encounter, state machine, triage, severity, owner, SLA, timeline.
4. Communication: acknowledgement and patient-safe tracking.
5. Management queue and complaint detail.
6. Only after those are stable, add management issue detection, recommendations, action plan, and evaluation.

- End of Q-VOICE Developer Handoff Specification -